

1. understand how the properties of solid materials depend on how the particles they are made from are arranged and held together;
2. relate the strength of the forces between the particles to the amount of energy needed for them to break out of the solid structure, and to the temperature at which the solid melts;
3. understand how modifications in polymers produce changes to their properties (see C2.1), to include modifications such as:
 - increased chain length;
 - cross-linking;
 - the use of plasticizers;
 - **increased crystallinity.**